

Nikhil Madaravena

Software Engineer | Full-Stack & Systems Engineering

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Professional Summary

Results-oriented Software Engineer specializing in backend systems, scalable full-stack applications, and performance-focused engineering. Experienced in building production-grade platforms using **Java Spring Boot, React.js, Node.js, PostgreSQL, and Rust** with strong foundations in **system design, operating systems, database architecture, and algorithms**.

Built scalable systems including multi-tenant SaaS platforms, high-performance in-memory databases, healthcare discovery systems, and real-time 3D visualization applications. Achieved measurable engineering impact including **40% latency reduction, 35% API performance improvement**, and optimized rendering pipelines sustaining **60 FPS**.

Recognized through **NASA Space Apps Challenge national-level selection, TCS NQT Programming excellence**, and competitive programming achievements.

Technical Skills

Languages	Java, Rust, C, Python, Go, JavaScript, TypeScript, SQL
Backend Engineering	Spring Boot, Spring Security, Hibernate, Node.js, Express.js, REST APIs, JWT Authentication, RBAC, Microservices
Frontend Engineering	React.js, Three.js, Tailwind CSS, Chart.js, Framer Motion, GSAP, HTML5, CSS3
Databases	PostgreSQL, MySQL, MongoDB, Query Optimization, Schema Design, Indexing
Systems & Architecture	Data Structures & Algorithms, OOP, Concurrent Programming, System Design, Database Internals
DevOps & Tools	Git, GitHub, Docker, Jenkins, Postman, Netlify, Vercel, Figma

Experience

Web Development Summer Intern May 2024 – Jul 2024
Kakatiya Institute of Technology and Science (KITSW) Warangal, India

- Developed the Institute Document Retrieval System (IDRS), a centralized full-stack document management platform serving administrative workflows.
- Reduced manual document retrieval overhead by **50%** through digitization and automated categorization pipelines.
- Implemented secure **Role-Based Access Control (RBAC)** with granular permission management.
- Optimized relational database queries reducing average document retrieval time from **4.2s to under 1.5s**.
- Built scalable REST APIs and integrated React.js frontend workflows for real-time document operations.

Projects

OpenBlog — Multi-User Blogging Platform *Java, Spring Boot, PostgreSQL, React.js*

- Developed a production-grade blogging platform supporting authentication, RBAC, nested comments, and secure CRUD workflows.
- Designed **20+ RESTful APIs** using layered Spring Boot architecture with JWT-based authentication and centralized exception handling.
- Optimized Hibernate query performance and lazy-loading strategies, reducing API response latency by **35%**.
- Implemented scalable backend architecture following Controller-Service-Repository design principles.

Kivo — High-Performance In-Memory Database *Rust, Concurrent Programming*

- Built a high-performance in-memory key-value database using custom hash structures and open addressing.
- Reduced lookup latency by **40%** through memory-efficient layouts and cache-aware optimization techniques.

- Implemented concurrent-safe operations using Rust ownership principles without garbage collection overhead.
- Added TTL expiration, eviction policies, and extensible command interfaces inspired by Redis-style architecture.

Makko — Multi-Tenant SaaS Website Builder

React.js, TypeScript, Node.js, PostgreSQL

- Engineered a multi-tenant SaaS platform enabling drag-and-drop website generation with real-time rendering.
- Designed secure tenant-isolated authentication and workspace management using JWT authorization.
- Built scalable backend APIs maintaining **sub-200ms** response times during concurrent load testing.
- Developed modular component architecture supporting dynamic widget generation and extensibility.

ClinicPing — Smart Healthcare Discovery Platform

React.js, Node.js, Leaflet.js

- Developed a location-aware healthcare platform for hospital discovery and appointment management.
- Integrated geospatial visualization using Leaflet.js and interactive 3D globe rendering.
- Built automated fallback appointment allocation logic improving booking success rates by **30%**.
- Designed REST APIs for live facility availability tracking and scheduling.

NeoVision — 3D Near-Earth Object Visualization System

React.js, Three.js, NASA APIs

- Built a real-time 3D visualization platform rendering NASA near-earth object data at stable **60 FPS**.
- Improved rendering efficiency by **35%** using frustum culling, geometry instancing, and LOD optimization.
- Integrated live NASA orbital datasets with smooth interpolation and interactive camera controls.
- Deployed production-ready application using automated CI/CD workflows on Netlify.

Achievements

- **NASA Space Apps Challenge** — National-Level Nominee at Chandigarh University.
- **TCS NQT** — Achieved **100% score** in Programming section.
- **Winner** — **CODE4KITSW Competitive Programming Championship**.
- **HackerRank** — Gold Badge in C Programming and Silver Badge in Java.
- **Google Cloud Computing Foundations Certified**.

Education

Bachelor of Technology in Computer Science and Engineering

2023 – 2027

Kakatiya Institute of Technology and Science (KITSW)

Warangal, India

Relevant Coursework: Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, OOP, System Design, Compiler Design

Leadership & Activities

Executive Member — Engineering & Technology Society

2023 – Present

KITSW Technical Club

Warangal, India

- Organized and coordinated inter-collegiate coding competitions and technical workshops.
- Mentored junior students in Data Structures, Algorithms, and software engineering fundamentals.
- Contributed to technical event planning, outreach, and peer-learning initiatives.